**TABLE 61-5 FDA Category I Monographed Sunscreen Ingredients**

| INGREDIENTS | MAXIMUM CONCENTRATION, % |
|---|---|
| *p*-Aminobenzoic acid (PABA) | 15 |
| Avobenzone | 3 |
| Cinoxate | 3 |
| Dioxybenzone (benzophenone-8) | 3 |
| Ecamsule | 15 |
| Homosalate | 15 |
| Methyl anthranilate | 5 |
| Octocrylene | 10 |
| Octyl methoxycinnamate | 7.5 |
| Octyl salicylate | 5 |
| Oxybenzone (benzophenone-3) | 6 |
| Padimate O (octyl dimethyl PABA) | 8 |
| Phenylbenzimidazole sulfonic acid | 4 |
| Sulisobenzone (benzophenone-4) | 10 |
| Titanium dioxide | 25 |
| Trolamine salicylate | 12 |
| Zinc oxide | 25 |

*Abbreviation*: FDA, U.S. Food and Drug Administration.

coverage, and water or sweat resistance are recommended for adequate sun protection.

Some degree of photoprotection can be achieved by limiting the time of sun exposure during the day. Since a large part of an individual's total lifetime sun exposure may occur by age 18, it is important to educate parents and young children about the hazards of sunlight. Eliminating exposure at midday will substantially reduce lifetime UVR exposure.

## PHOTOTHERAPY AND PHOTOCHEMOTHERAPY

UVR can be used therapeutically. The administration of UV-B alone or in combination with topically applied agents can induce remissions of many dermatologic diseases, including psoriasis, atopic dermatitis, and vitiligo. In particular, narrow-band UV-B treatments (with fluorescent bulbs emitting radiation at ~311 nm) have enhanced efficacy over that obtained with broad-band UV-B in the treatment of psoriasis.

Photochemotherapy in which topically applied or systemically administered psoralens are combined with UV-A (PUVA) is effective in treating psoriasis and the early stages of cutaneous T-cell lymphoma and vitiligo. Psoralens are tricyclic furocoumarins that, when intercalated into DNA and exposed to UV-A, form adducts with pyrimidine bases and eventually form DNA cross-links. These structural changes are thought to decrease DNA synthesis and to be related to the amelioration of psoriasis. Why PUVA photochemotherapy is effective in cutaneous T-cell lymphoma is only partially understood, but it has been shown to induce apoptosis of atypical T lymphocyte populations in the skin. Consequently, direct treatment of circulating atypical lymphocytes by extracorporeal photochemotherapy (photopheresis) has been used in Sézary syndrome as well as in other severe systemic diseases with circulating atypical lymphocytes, such as graft-versus-host disease.

In addition to its effects on DNA, PUVA photochemotherapy stimulates epidermal thickening and melanin synthesis; the latter property, together with its anti-inflammatory effects, provides the rationale for use of PUVA in the depigmenting disease vitiligo. Oral 8-methoxypsoralen and UV-A appear to be most effective in this regard, but as many as 100 treatments extending over 12–18 months may be required for satisfactory repigmentation.

Not surprisingly, the major side effects of long-term UV-B phototherapy and PUVA photochemotherapy mimic those seen in individuals with chronic sun exposure. Despite these risks, the therapeutic index of these modalities continues to be excellent. It is important to choose the most appropriate phototherapeutic approach for a specific dermatologic disease. For example, narrow-band UV-B has been reported in several studies to be as effective as PUVA photochemotherapy in the treatment of psoriasis but to pose a lower risk of skin cancer development than PUVA.

## FURTHER READING


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# Section 9 Hematologic Alterations

## 62 Interpreting Peripheral Blood Smears

**Dan L. Longo**

Some of the relevant findings in peripheral blood, enlarged lymph nodes, and bone marrow are illustrated in this chapter. Systematic histologic examination of the bone marrow and lymph nodes is beyond the scope of a general medicine textbook. However, every internist should know how to examine a peripheral blood smear.

The examination of a peripheral blood smear is one of the most informative exercises a physician can perform. Although advances in automated technology have made the examination of a peripheral blood smear by a physician seem less important, the technology is not a completely satisfactory replacement for a blood smear interpretation by a trained medical professional who also knows the patient's clinical history, family history, social history, and physical findings. It is useful to ask the laboratory to generate a Wright's-stained peripheral blood smear and examine it.

The best place to examine blood cell morphology is the feathered edge of the blood smear where red cells lie in a single layer, side by side, just barely touching one another but not overlapping. The author's approach is to look at the smallest cellular elements, the platelets, first and work his way up in size to red cells and then white cells.

Using an oil immersion lens that magnifies the cells 100-fold, one counts the platelets in five to six fields, averages the number per field, and multiplies by 20,000 to get a rough estimate of the platelet count. The platelets are usually 1–2 μm in diameter and have a blue granulated appearance. There is usually 1 platelet for every 20 or so red cells. Of course, the automated counter is much more accurate, but gross disparities between the automated and manual counts should be assessed. Large platelets may be a sign of rapid platelet turnover, as young platelets are often larger than old ones; alternatively, certain rare inherited syndromes can produce large platelets. If the platelet count is low, the absence of large (young) platelets may be an indicator of marrow production problems. Platelet clumping visible on the smear

can be associated with falsely low automated platelet counts. Clumping may be caused by the anticoagulant into which the blood is drawn. Similarly, neutrophil fragmentation can be a source of falsely elevated automated platelet counts. The absence of platelet granules may be an artifact of the handling of the blood or may indicate marrow disease or a rare congenital anomaly, gray platelet syndrome. Elevated platelet counts usually signify a myeloproliferative disorder or a reaction to systemic inflammation.

Next one examines the red blood cells. One can gauge their size by comparing the red cell to the nucleus of a small lymphocyte. Both are normally about 8-μm wide. Red cells that are smaller than the small lymphocyte nucleus may be microcytic; those larger than the small lymphocyte nucleus may be macrocytic. Macrocytic cells also tend to be more oval than spherical in shape and are sometimes called macroovalocytes. The automated mean corpuscular volume (MCV) can assist in making a classification. However, some patients may have both iron and vitamin $B_{12}$ deficiency, which will produce an MCV in the normal range but wide variation in red cell size. When the red cells vary greatly in size, *anisocytosis* is said to be present. When the red cells vary greatly in shape, *poikilocytosis* is said to be present. The electronic cell counter provides an independent assessment of variability in red cell size. It measures the range of red cell volumes and reports the results as "red cell distribution width" (RDW). This value is calculated from the MCV; thus, cell width is not being measured but cell volume is. The term is derived from the curve displaying the frequency of cells at each volume, also called the distribution. The width of red cell volume distribution curve is what determines the RDW. The RDW is calculated as follows: RDW = (standard deviation of MCV ÷ mean MCV) × 100. In the presence of morphologic anisocytosis, RDW (normally 11–14%) increases to 15–18%. The RDW is useful in at least two clinical settings. In patients with microcytic anemia, the differential diagnosis is generally between iron deficiency and thalassemia. In thalassemia, the small red cells are generally of uniform size with a normal small RDW. In iron deficiency, the size variability and the RDW are large. In addition, a large RDW can suggest a dimorphic anemia when a chronic atrophic gastritis can produce both vitamin $B_{12}$ malabsorption to produce macrocytic anemia and blood loss to produce iron deficiency. In such settings, RDW is also large. An elevated RDW also has been reported as a risk factor for all-cause mortality in population-based studies, a finding that is unexplained currently.

After red cell size is assessed, one examines the hemoglobin content of the cells. They are either normal in color (*normochromic*) or pale in color (*hypochromic*). They are never "hyperchromic." If more than the normal amount of hemoglobin is made, the cells get larger—they do not become darker. In addition to hemoglobin content, the red cells are examined for inclusions. Red cell inclusions are the following:

1. *Basophilic stippling*—diffuse fine or coarse blue dots in the red cell usually representing RNA residue—especially common in lead poisoning
2. *Howell-Jolly bodies*—dense blue circular inclusions that represent nuclear remnants—their presence implies defective splenic function
3. *Nuclei*—red cells may be released or pushed out of the marrow prematurely before nuclear extrusion—often implies a myelophthisic process or a vigorous narrow response to anemia, usually hemolytic anemia
4. *Parasites*—red cell parasites include malaria and babesia **(Chap. A6)**
5. *Polychromatophilia*—the red cell cytoplasm has a bluish hue, reflecting the persistence of ribosomes still actively making hemoglobin in a young red cell

Vital stains are necessary to see precipitated hemoglobin called *Heinz bodies*.

Red cells can take on a variety of different shapes. All abnormally shaped red cells are *poikilocytes*. Small red cells without the central pallor are *spherocytes*; they can be seen in hereditary spherocytosis, hemolytic anemias of other causes, and clostridial sepsis. *Dacrocytes* are teardrop-shaped cells that can be seen in hemolytic anemias, severe iron deficiency, thalassemias, myelofibrosis, and myelodysplastic syndromes. *Schistocytes* are helmet-shaped cells that reflect microangiopathic hemolytic anemia or fragmentation on an artificial heart valve. *Echinocytes* are spiculated red cells with the spikes evenly spaced; they can represent an artifact of abnormal drying of the blood smear or reflect changes in stored blood. They also can be seen in renal failure and malnutrition and are often reversible. *Acanthocytes* are spiculated red cells with the spikes irregularly distributed. This process tends to be irreversible and reflects underlying renal disease, abetalipoproteinemia, or splenectomy. *Elliptocytes* are elliptical-shaped red cells that can reflect an inherited defect in the red cell membrane, but they also are seen in iron deficiency, myelodysplastic syndromes, megaloblastic anemia, and thalassemias. *Stomatocytes* are red cells in which the area of central pallor takes on the morphology of a slit instead of the usual round shape. Stomatocytes can indicate an inherited red cell membrane defect and also can be seen in alcoholism. *Target cells* have an area of central pallor that contains a dense center, or bull's eye. These cells are seen classically in thalassemia, but they are also present in iron deficiency, cholestatic liver disease, and some hemoglobinopathies. They also can be generated artifactually by improper slide making.

One last feature of the red cells to assess before moving to the white blood cells is the distribution of the red cells on the smear. In most individuals, the cells lie side by side in a single layer. Some patients have red cell clumping (called *agglutination*) in which the red cells pile upon one another; it is seen in certain paraproteinemias and autoimmune hemolytic anemias. Another abnormal distribution involves red cells lying in single cell rows on top of one another like stacks of coins. This is called *rouleaux formation* and reflects abnormal serum protein levels.

Finally, one examines the white blood cells. Three types of granulocytes are usually present: neutrophils, eosinophils, and basophils, in decreasing frequency. Neutrophils are generally the most abundant white cell. They are round, are 10–14 μm wide, and contain a lobulated nucleus with two to five lobes connected by a thin chromatin thread. Bands are immature neutrophils that have not completed nuclear condensation and have a U-shaped nucleus. Bands reflect a left shift in neutrophil maturation in an effort to make more cells more rapidly. Neutrophils can provide clues to a variety of conditions. Vacuolated neutrophils may be a sign of bacterial sepsis. The presence of 1- to 2-μm blue cytoplasmic inclusions, called *Döhle bodies*, can reflect infections, burns, or other inflammatory states. If the neutrophil granules are larger than normal and stain a darker blue, "toxic granulations" are said to be present, and they also suggest a systemic inflammation. The presence of neutrophils with more than five nuclear lobes suggests megaloblastic anemia. Large misshapen granules may reflect the inherited Chédiak-Higashi syndrome.

Eosinophils are slightly larger than neutrophils, have bilobed nuclei, and contain large red granules. Diseases of eosinophils are associated with too many of them rather than any morphologic or qualitative change. They normally total less than one-thirtieth the number of neutrophils. Basophils are even more rare than eosinophils in the blood. They have large dark blue granules and may be increased as part of chronic myeloid leukemia.

Lymphocytes can be present in several morphologic forms. Most common in healthy individuals are small lymphocytes with a small dark nucleus and scarce cytoplasm. In the presence of viral infections, more of the lymphocytes are larger, about the size of neutrophils, with abundant cytoplasm and a less condensed nuclear chromatin. These cells are called *reactive lymphocytes*. About 1% of lymphocytes are larger and contain blue granules in a light blue cytoplasm; they are called *large granular lymphocytes*. In chronic lymphoid leukemia, the small lymphocytes are increased in number, and many of them are ruptured in making the blood smear, leaving a smudge of nuclear material without a surrounding cytoplasm or cell membrane; they are called *smudge cells* and are rare in the absence of chronic lymphoid leukemia.

Monocytes are the largest white blood cells, ranging from 15 to 22 μm in diameter. The nucleus can take on a variety of shapes but usually appears to be folded; the cytoplasm is gray.

Abnormal cells may appear in the blood. Most often, the abnormal cells originate from neoplasms of bone marrow–derived cells, including lymphoid cells, myeloid cells, and occasionally red cells. More rarely, other types of tumors can get access to the bloodstream, and rare

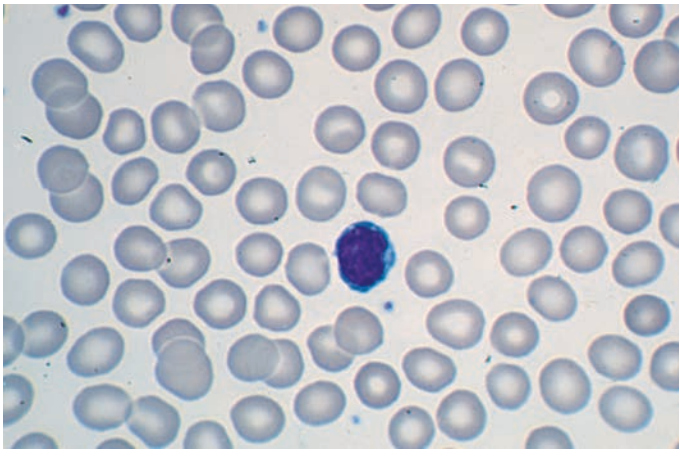


**FIGURE 62-1 Normal peripheral blood smear.** Small lymphocyte in center of field. Note that the diameter of the red blood cell is similar to the diameter of the small lymphocyte nucleus. *(Source: From M Lichtman et al (eds). Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault, Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

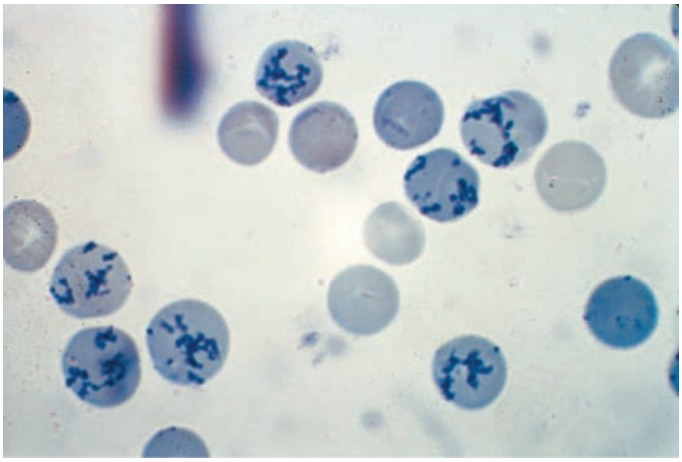


**FIGURE 62-2 Reticulocyte count preparation.** This new methylene blue–stained blood smear shows large numbers of heavily stained reticulocytes (the cells containing the dark blue–staining RNA precipitates). *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

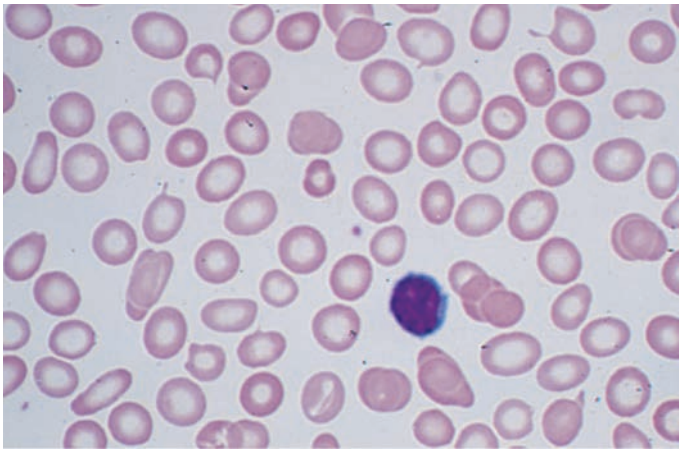


**FIGURE 62-3 Hypochromic microcytic anemia of iron deficiency.** Small lymphocyte in field helps assess the red blood cell size. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

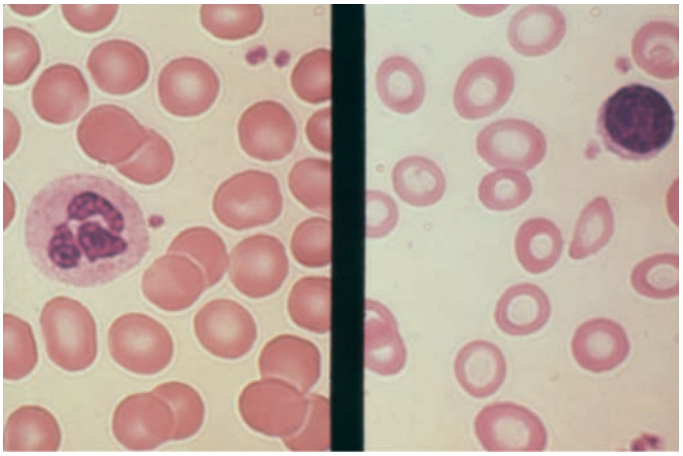


**FIGURE 62-4 Iron deficiency anemia next to normal red blood cells.** Microcytes (*right panel*) are smaller than normal red blood cells (cell diameter <7 μm) and may or may not be poorly hemoglobinized (hypochromic). *(Source: From M Lichtman et al (eds). Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault, Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

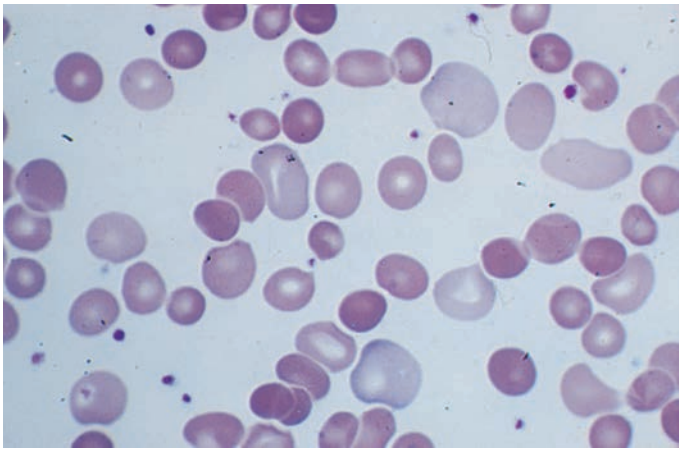


**FIGURE 62-5 Polychromatophilia.** Note large red cells with light purple coloring. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

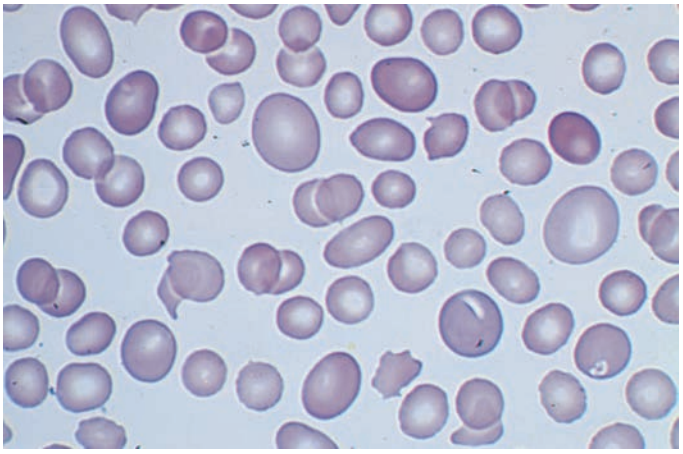


**FIGURE 62-6 Macrocytosis.** These cells are both larger than normal (mean corpuscular volume >100) and somewhat oval in shape. Some morphologists call these cells macroovalocytes. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

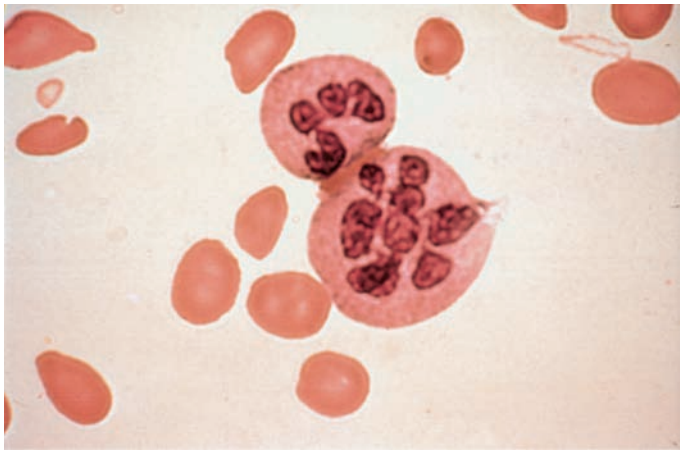


**FIGURE 62-7 Hypersegmented neutrophils.** Hypersegmented neutrophils (multilobed polymorphonuclear leukocytes) are larger than normal neutrophils with five or more segmented nuclear lobes. They are commonly seen with folic acid or vitamin $B_{12}$ deficiency. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

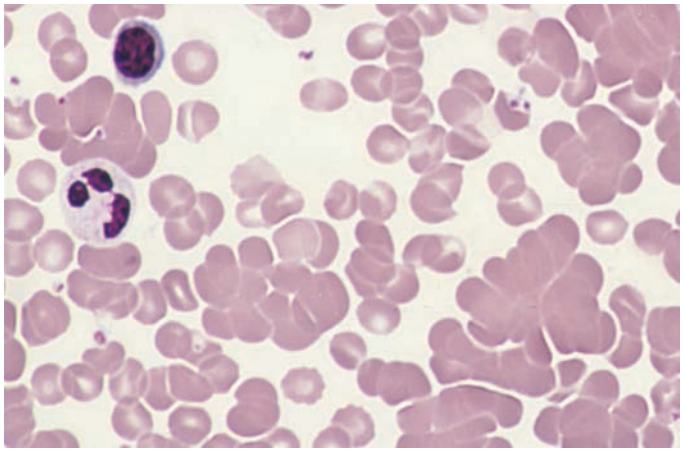


**FIGURE 62-10 Red cell agglutination.** Small lymphocyte and segmented neutrophil in upper left center. Note irregular collections of aggregated red cells. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

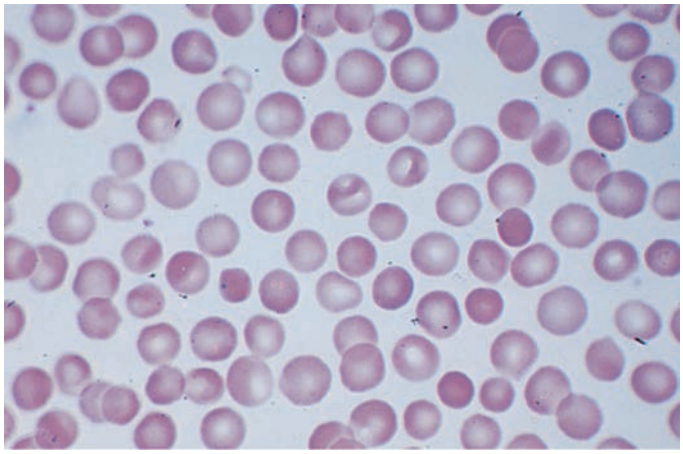


**FIGURE 62-8 Spherocytosis.** Note small hyperchromatic cells without the usual clear area in the center. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

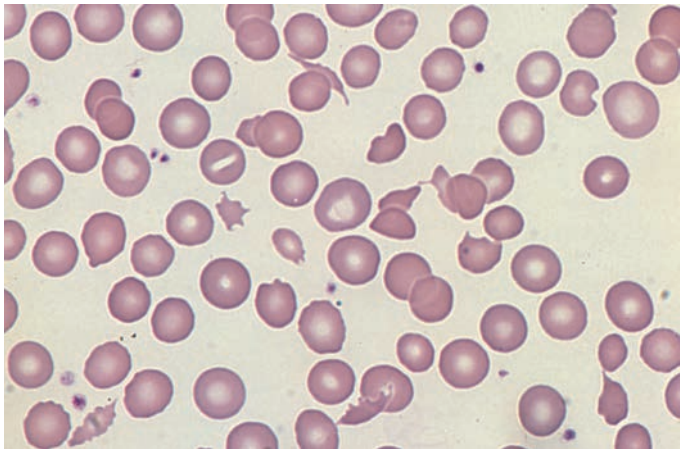


**FIGURE 62-11 Fragmented red cells.** Heart valve hemolysis. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

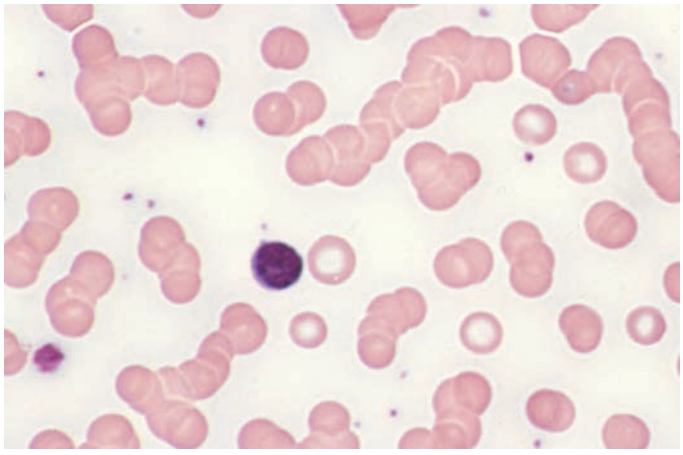


**FIGURE 62-9 Rouleaux formation.** Small lymphocyte in center of field. These red cells align themselves in stacks and are related to increased serum protein levels. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

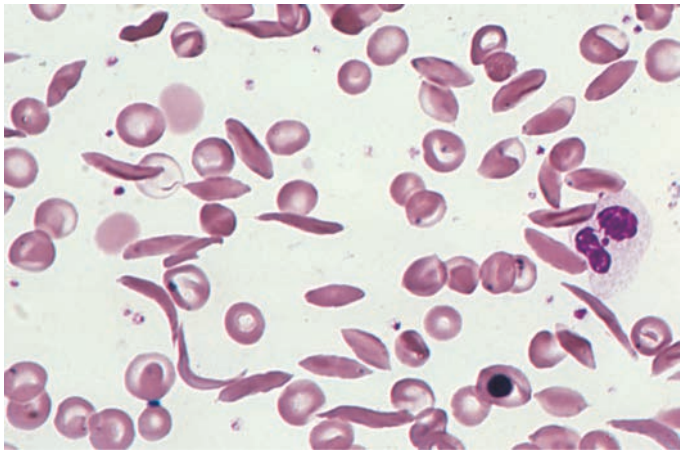


**FIGURE 62-12 Sickle cells.** Homozygous sickle cell disease. A nucleated red cell and neutrophil are also in the field. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

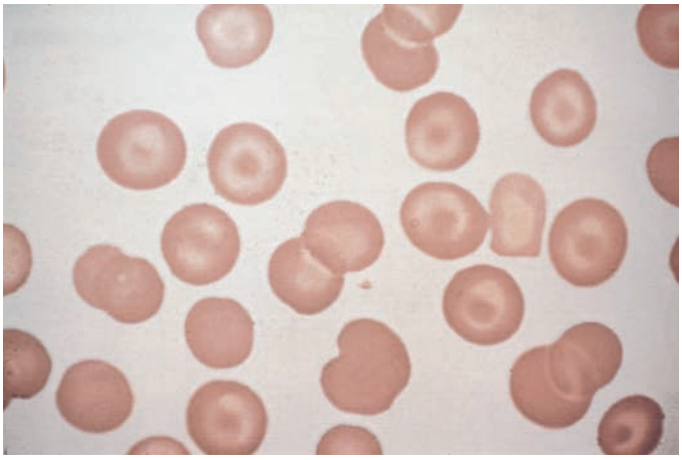


**FIGURE 62-13 Target cells.** Target cells are recognized by the bull's-eye appearance of the cell. Small numbers of target cells are seen with liver disease and thalassemia. Larger numbers are typical of hemoglobin C disease. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

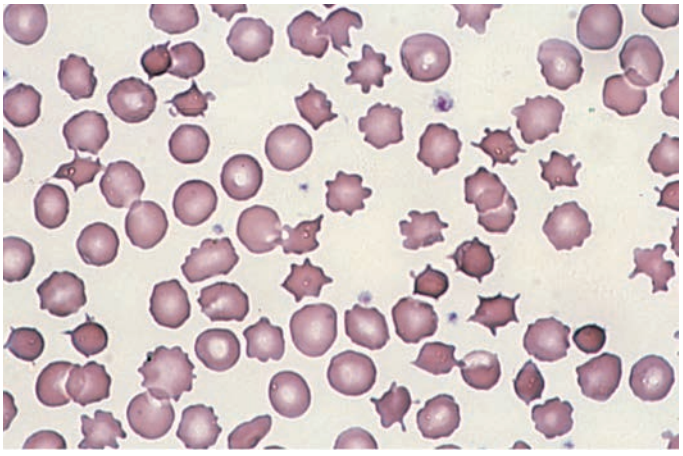


**FIGURE 62-16 Acanthocytosis.** Spiculated red cells are of two types: *acanthocytes* are contracted dense cells with irregular membrane projections that vary in length and width; *echinocytes* have small, uniform, and evenly spaced membrane projections. Acanthocytes are present in severe liver disease, in patients with abetalipoproteinemia, and in rare patients with McLeod blood group. Echinocytes are found in patients with severe uremia, in glycolytic red cell enzyme defects, and in microangiopathic hemolytic anemia. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

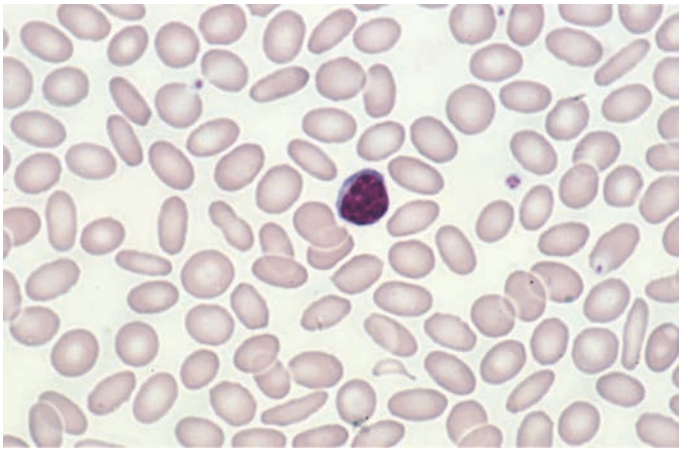


**FIGURE 62-14 Elliptocytosis.** Small lymphocyte in center of field. Elliptical shape of red cells related to weakened membrane structure, usually due to mutations in spectrin. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

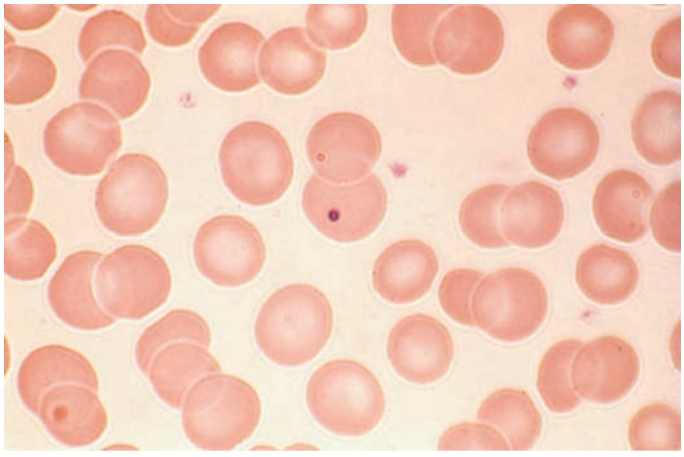


**FIGURE 62-17 Howell-Jolly bodies.** Howell-Jolly bodies are tiny nuclear remnants that normally are removed by the spleen. They appear in the blood after splenectomy (defect in removal) and with maturation/dysplastic disorders (excess production). *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

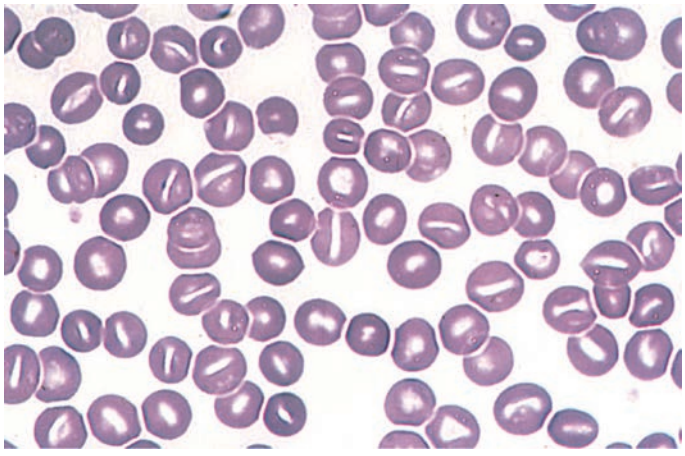


**FIGURE 62-15 Stomatocytosis.** Red cells characterized by a wide transverse slit or stoma. This often is seen as an artifact in a dehydrated blood smear. These cells can be seen in hemolytic anemias and in conditions in which the red cell is overhydrated or dehydrated. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

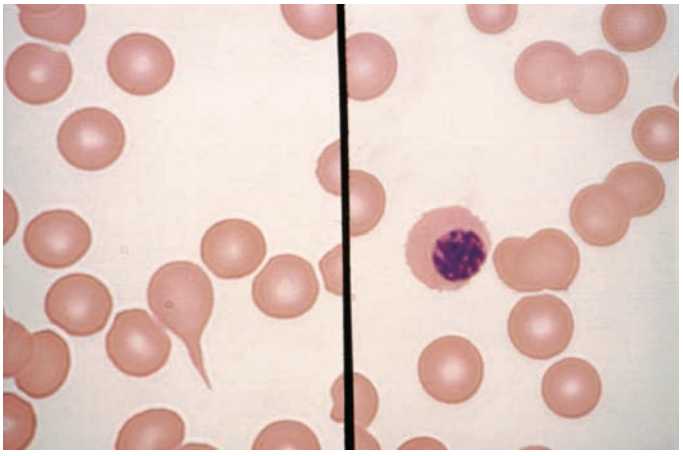


**FIGURE 62-18 Teardrop cells and nucleated red blood cells characteristic of myelofibrosis.** A teardrop-shaped red blood cell (*left panel*) and a nucleated red blood cell (*right panel*) as typically seen with myelofibrosis and extramedullary hematopoiesis. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

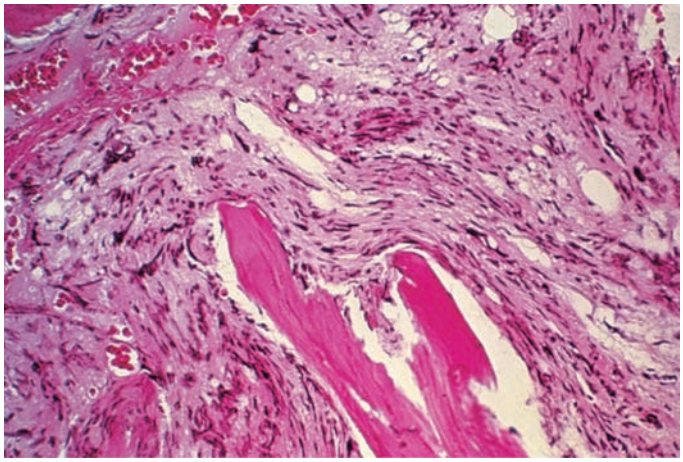


**FIGURE 62-19 Myelofibrosis of the bone marrow.** Total replacement of marrow precursors and fat cells by a dense infiltrate of reticulin fibers and collagen (hematoxylin and eosin stain). *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

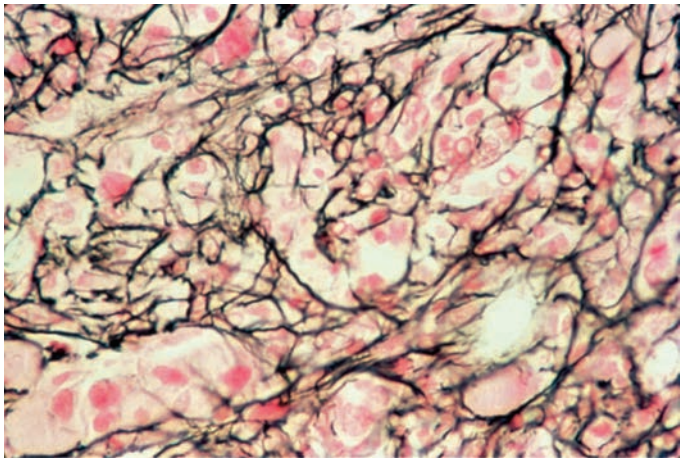


**FIGURE 62-20 Reticulin stain of marrow myelofibrosis.** Silver stain of a myelofibrotic marrow showing an increase in reticulin fibers (black-staining threads). *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

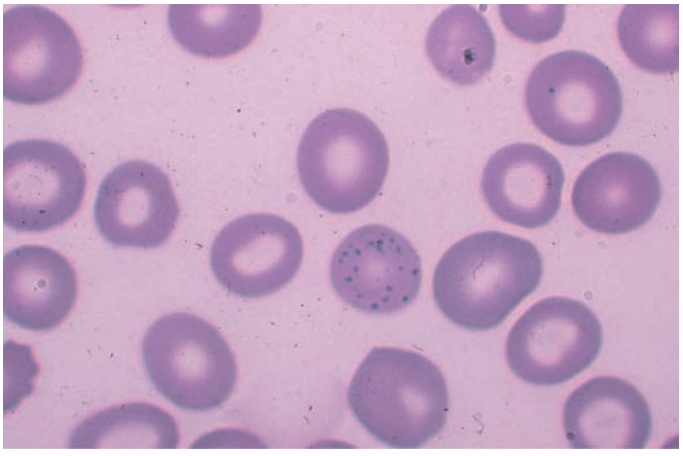


**FIGURE 62-21 Stippled red cell in lead poisoning.** Mild hypochromia. Coarsely stippled red cell. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

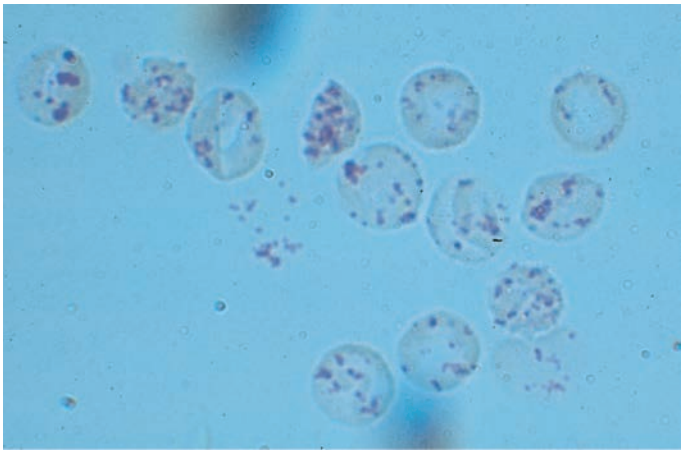


**FIGURE 62-22 Heinz bodies.** Blood mixed with hypotonic solution of crystal violet. The stained material is precipitates of denatured hemoglobin within cells. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

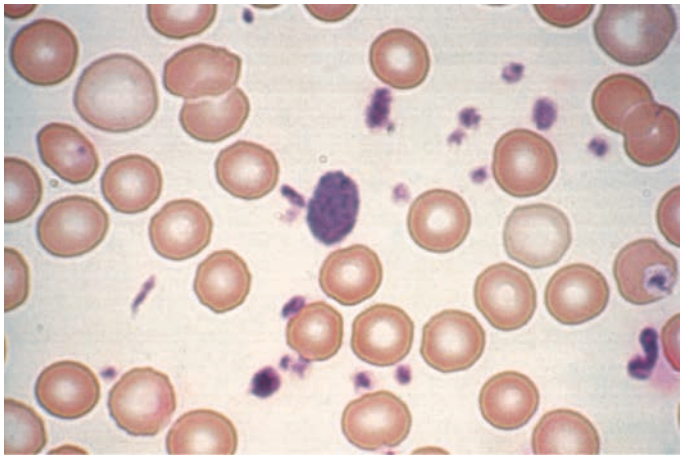


**FIGURE 62-23 Giant platelets.** Giant platelets, together with a marked increase in the platelet count, are seen in myeloproliferative disorders, especially primary thrombocythemia. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

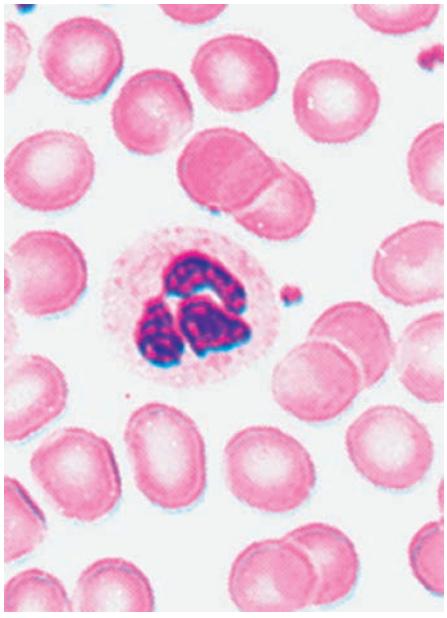


**FIGURE 62-24 Normal granulocytes.** The normal granulocyte has a segmented nucleus with heavy, clumped chromatin; fine neutrophilic granules are dispersed throughout the cytoplasm. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

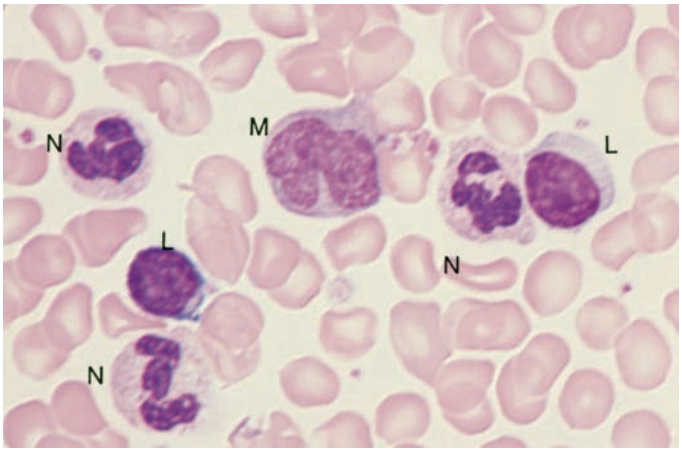



**FIGURE 62-25 Normal monocytes.** The film was prepared from the buffy coat of the blood from a normal donor. L, lymphocyte; M, monocyte; N, neutrophil. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

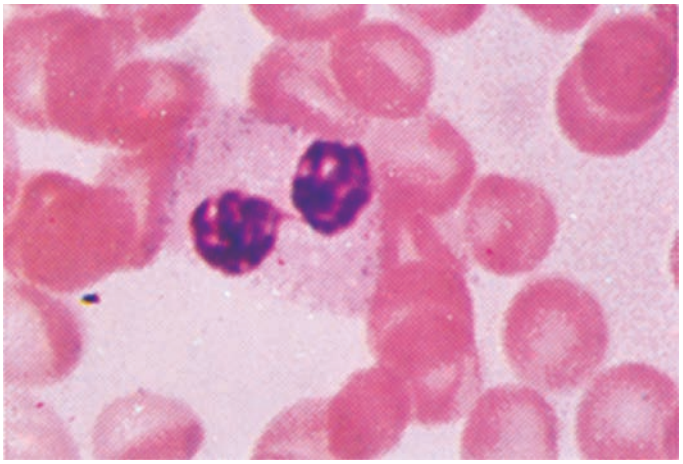


**FIGURE 62-28 Pelger-Hüet anomaly.** In this benign disorder, the majority of granulocytes are bilobed. The nucleus frequently has a spectacle-like, or "pince-nez," configuration. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

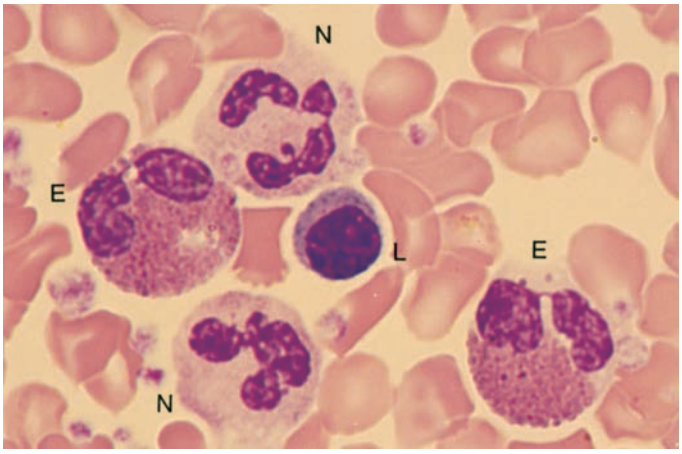



**FIGURE 62-26 Normal eosinophils.** The film was prepared from the buffy coat of the blood from a normal donor. E, eosinophil; L, lymphocyte; N, neutrophil. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

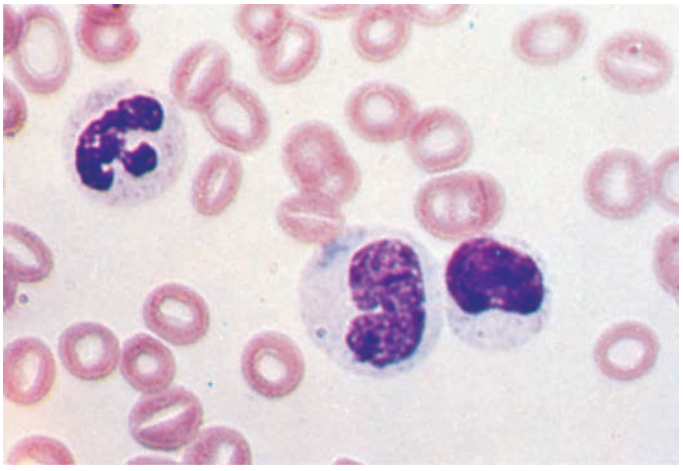


**FIGURE 62-29 Döhle body.** Neutrophil band with Döhle body. The neutrophil with a sausage-shaped nucleus in the center of the field is a band form. Döhle bodies are discrete, blue-staining nongranular areas found in the periphery of the cytoplasm of the neutrophil in infections and other toxic states. They represent aggregates of rough endoplasmic reticulum. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

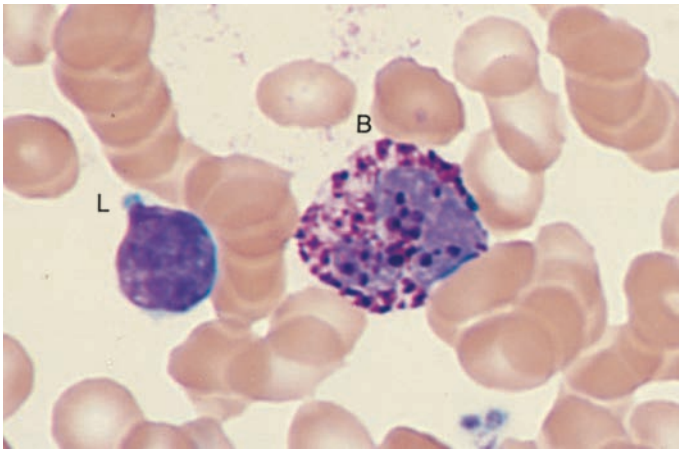



**FIGURE 62-27 Normal basophil.** The film was prepared from the buffy coat of the blood from a normal donor. B, basophil; L, lymphocyte. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

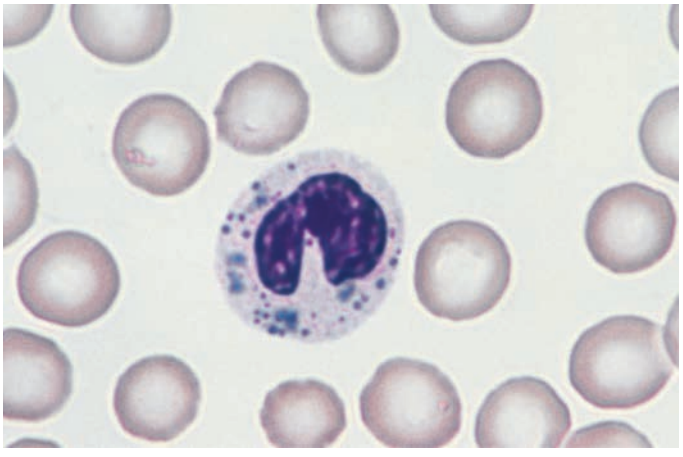


**FIGURE 62-30 Chédiak-Higashi disease.** Note giant granules in neutrophil. *(Source: From M Lichtman et al (eds): Williams Hematology, 7th ed. New York, McGraw-Hill, 2005; RS Hillman, KA Ault: Hematology in General Practice, 4th ed. New York, McGraw-Hill, 2005.)*

epithelial malignant cells may be identified. The chances of seeing such abnormal cells are increased by examining blood smears made from buffy coats, the layer of cells that is visible on top of sedimenting red cells when blood is left in the test tube for an hour. Smears made from finger sticks may include rare endothelial cells.

**Acknowledgment**

*Figures in this chapter were borrowed from Williams Hematology, 7th edition, M Lichtman et al (eds). New York, McGraw-Hill, 2005; Hematology in General Practice, 4th edition, RS Hillman, KA Ault. New York, McGraw-Hill, 2005.*

# 63 Anemia and Polycythemia

**John W. Adamson, Dan L. Longo**

## HEMATOPOIESIS AND THE PHYSIOLOGIC BASIS OF RED CELL PRODUCTION

*Hematopoiesis* is the process by which the formed elements of blood are produced. The process is regulated through a series of steps beginning with the hematopoietic stem cell. Stem cells are capable of producing red cells, all classes of granulocytes, monocytes, platelets, and the cells of the immune system. The precise molecular mechanism by which the stem cell becomes committed to a given lineage is not fully defined. However, experiments in mice suggest that erythroid cells come from a common erythroid/megakaryocyte progenitor that does not develop in the absence of expression of the GATA-1 and FOG-1 (friend of GATA-1) transcription factors **(Chap. 96)**. Following lineage commitment, hematopoietic progenitor and precursor cells come increasingly under the regulatory influence of growth factors and hormones. For red cell production, erythropoietin (EPO) is the primary regulatory hormone. EPO is required for the maintenance of committed erythroid progenitor cells that, in the absence of the hormone, undergo programmed cell death (*apoptosis*). The regulated process of red cell production is *erythropoiesis*, and its key elements are illustrated in **Fig. 63-1**.

In the bone marrow, the first morphologically recognizable erythroid precursor is the pronormoblast. This cell can undergo four to five cell divisions, which result in the production of 16–32 mature red cells. With increased EPO production, or the administration of EPO as a drug, early progenitor cell numbers are amplified and, in turn, give rise to increased numbers of erythrocytes. The regulation of EPO production itself is linked to tissue oxygenation.

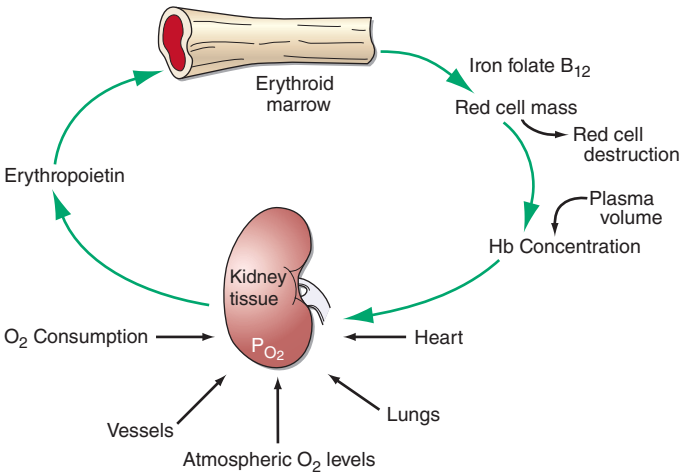



**FIGURE 63-1 The physiologic regulation of red cell production by tissue oxygen tension.** Hb, hemoglobin.

In mammals, $O_2$ is transported to tissues bound to the hemoglobin contained within circulating red cells. The mature red cell is 8 μm in diameter, anucleate, discoid in shape, and extremely pliable in order to traverse the microcirculation successfully; its membrane integrity is maintained by the intracellular generation of ATP. Normal red cell production results in the daily replacement of 0.8–1% of all circulating red cells in the body, since the average red cell lives 100–120 days. The organ responsible for red cell production is called the *erythron*. The erythron is a dynamic organ made up of a rapidly proliferating pool of marrow erythroid precursor cells and a large mass of mature circulating red blood cells. The size of the red cell mass reflects the balance of red cell production and destruction. The physiologic basis of red cell production and destruction provides an understanding of the mechanisms that can lead to anemia.

The physiologic regulator of red cell production, the glycoprotein hormone EPO, is produced and released by peritubular capillary lining cells within the kidney. These cells are highly specialized epithelial-like cells. A small amount of EPO is produced by hepatocytes. The fundamental stimulus for EPO production is the availability of $O_2$ for tissue metabolic needs. Key to EPO gene regulation is hypoxia-inducible factor (HIF)-1α. In the presence of $O_2$, HIF-1α is hydroxylated at a key proline, allowing HIF-1α to be ubiquitinated and degraded via the proteasome pathway. If $O_2$ becomes limiting, this critical hydroxylation step does not occur, allowing HIF-1α to partner with other proteins, translocate to the nucleus, and upregulate the expression of the EPO gene, among others.

Impaired $O_2$ delivery to the kidney can result from a decreased red cell mass (*anemia*), impaired $O_2$ loading of the hemoglobin molecule or a high $O_2$ affinity mutant hemoglobin (*hypoxemia*), or, rarely, impaired blood flow to the kidney (e.g., renal artery stenosis). EPO governs the day-to-day production of red cells, and ambient levels of the hormone can be measured in the plasma by sensitive immunoassays—the normal level being 10–25 U/L. When the hemoglobin concentration falls below 100–120 g/L (10–12 g/dL), plasma EPO levels increase in proportion to the severity of the anemia **(Fig. 63-2)**. In circulation, EPO has a half-clearance time of 6–9 h. EPO acts by binding to specific receptors on the surface of marrow erythroid precursors, inducing them to proliferate and to mature. With EPO stimulation, red cell production can increase four- to fivefold within a 1- to 2-week period, but only in the presence of adequate nutrients, especially iron. The functional capacity of the erythron, therefore, requires normal renal production of EPO, a functioning erythroid marrow, and an adequate supply of substrates for hemoglobin synthesis. A defect in any of these key components can lead to anemia. Generally, anemia is recognized in the laboratory when a patient's hemoglobin level or hematocrit is reduced below an expected value (the normal range).

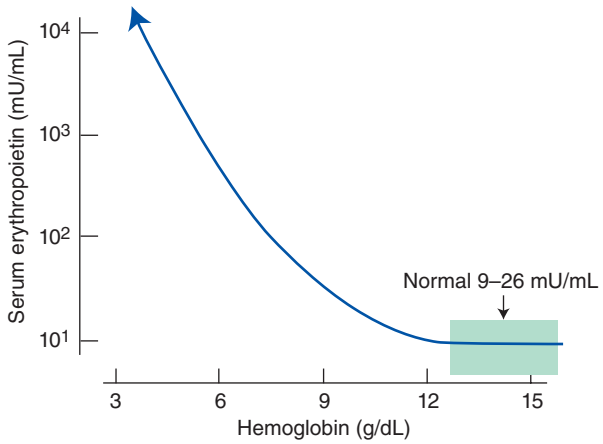



**FIGURE 63-2 Erythropoietin (EPO) levels in response to anemia.** When the hemoglobin level falls to 120 g/L (12 g/dL), plasma EPO levels increase logarithmically. In the presence of chronic kidney disease or chronic inflammation, EPO levels are typically lower than expected for the degree of anemia. As individuals age, the level of EPO needed to sustain normal hemoglobin levels appears to increase. *(Reproduced with permission from RS Hillman et al: Hematology in Clinical Practice, 5th ed. New York, McGraw-Hill, 2010.)*